Trainings

General Information

This procedure describes coolant specifications and service requirements for engine coolant. For more exhaustive information on coolant specifications and service requirements for engine coolant. Refer to Fluids for Cummins® Products Service Manual 5411406.

Coolant Specifications

Cummins Inc. requires the use of a coolant meeting the appropriate Cummins® Engineering Standard (CES) for the engine. Table 1 below shows the approved CES for various engine models and applications of the engine.

A list of coolants that meet the appropriate CES for your engine can be obtained from a local Cummins® distributor or can be found on QuickServe® Online (QSOL) at:

Note : https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/coolant_registration.html (https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/coolant_registration.html)

Table 1, Cummins® Engineering Standards (CES) for Coolant	
Engine Family	**CES**
Diesel ISV	CES 14636
Diesel 12 L Diesel 15 L On-road Diesel ISL Diesel QSB and ISB	CES 14603 CES 14439
All Other Engines	CES 14603

- For new equipment, check with the original equipment manufacturer (OEM) on the type of coolant used for the first fill. This will assist in understanding how to maintain the coolant properly.
- Cummins Inc. recommends a premixed, heavy-duty coolant when filling the cooling system.
  - See Fluids for Cummins ™ Products Service Manual, Bulletin 5411406. Refer to Procedure 379-005 (/qs3/pubsys2/xml/en/procedures/00/00-379-005.html) in Section 5, if diluting coolant from concentrate.
- Customers with marine or arctic applications should reference Fluids for Cummins ™ Products Service Manual, Bulletin 5411406. Refer to Procedure 379-004 (/qs3/pubsys2/xml/en/procedures/00/00-379-004.html) in Section 5 for information on coolant recommendations and specifications specific to those applications.

Coolant Service

Topping off the cooling system with an appropriate pre-diluted or 50/50 coolant is critical for maintaining additive concentrations and freeze protection. In general, different coolants should not be mixed. **Never** top off the cooling system with water **only**.

Coolant should be checked for: glycol concentration, SCA level and coolant replacement limits at different intervals depending on the type of coolant used. Table 2 below indicates minimum-required test intervals depending on coolant type:

Table 2, Test Intervals for Fleetguard® Coolants

Test	Conventional	Hybrid	OAT	Minimum Required Service Tool
Glycol[1]	Twice per year	Twice per year	Twice per year	CC2602
SCA Level[1]	Every lubricating oil drain	Shortest of: 241,401 km [150,000 miles], 4000 hours, or 1 year	**Not** required	CC2602
Coolant Replacement Limits Check	Shortest of: 241,401 km [150,000 miles], 4000 hours, or 1 year	Shortest of: 241,401 km [150,000 miles], 4000 hours, or 1 year	Shortest of: 482,803 km [300,000 miles], 6000 hours, or 1 year	CC2718

Refer to Procedure 379-007 (/qs3/pubsys2/xml/en/procedures/00/00-379-007.html) in Section 5, for a complete list of all applicable coolant testing tools.

Cummins Inc. recommends targeting a glycol/antifreeze concentration of 50 percent by volume. Glycol concentration must be maintained according to Table 3.

Table 3, Glycol Concentration

Concentration Requirement	Out of Range Actions
40 to 60 Percent	Drain a portion of coolant and replace with 50/50 coolant to get back in the 40 to 60 percent glycol range. Under-concentration of glycol occurs when filling or topping off with straight water or over-diluted coolant. Over-concentration of glycol occurs in high heat situations where there is some water evaporation from the cooling system or when filling or topping off with improperly diluted coolant.

SCA **must** be maintained according to Table 4.

Table 4, Guidelines for SCA Addition	
SCA Concentration	**Recommendation**
Less than 0.3 units/L [1.2 units/U.S. gal]	Add 0.15 L [5 oz.] of liquid SCA per 3.8 L [1 U.S. gal] of coolant. Install the correct chemical coolant filter, if applicable. For large cooling systems, it may be necessary to use liquid SCA in conjunction with a chemical filter to replenish SCA.
0.3 units/L [1.2 units/U.S. gal] to 0.8 units/L [3.0 units/U.S. gal]	Install the correct chemical coolant filter or add the correct liquid SCA and install a chemical-free coolant filter.
Greater than 0.8 units/L [3.0 units/U.S. gal] to 1.3 units/L [5.0 units/U.S. gal]	Install the correct chemical-free coolant filter, if applicable. Do **not** add liquid SCA. Test the SCA level at each oil change until the SCA drops below 0.8 units/L [3.0 units/U.S. gal]. Drain and replace the coolant if overheating occurs or if cooling system corrosion is detected by visual changes to coolant's appearance or by laboratory testing.

Table 4 is only for customers using coolants with conventional / hybrid coolants that use SCA. Organic acid technology (OAT)/Extended Life coolants use high levels of salts of organic acids for cylinder liner pitting protection and do **not** use nitrite and or molybdate. Therefore, these coolants do **not** have SCA numbers. Additionally, OAT additives deplete much slower than conventional additives and as a result OAT coolants do **not** require addition of SCA products.

Replace the coolant **only** if the replacement limits are exceeded. Coolant replacement limits are listed in Table 5:

Table 5, Coolant Replacement Limits		
Contaminant	**Allowable Level**	**Out of Range Actions**
Sulfate (SO_4)	1500 ppm, maximum	Drain and replace with new coolant
Chloride (Cl-)+, Fluoride (F-)+, Bromide (Br-)	100 ppm, combined maximum	
Glycolate	3000 ppm, maximum	
pH	6.5 minimum / 10.5 maximum Note: Some Cummins® Engineering Standards approved coolants start at pH levels of 10,5 and will decrease naturally with time.	
Oil or fuel contamination	Coolant **must** be free of these contaminants	Drain, clean, and replace with new coolant. Refer to Procedure 379-009 in Section 5.
Grease, solder bloom, silica gel, rust, or scaling	Coolant **must** be free of these contaminants	

Refer to Procedure 379-007 (/qs3/pubsys2/xml/en/procedures/00/00-379-007.html) in Section 5, for additional information on testing methods. Refer to Procedure 379-008 (/qs3/pubsys2/xml/en/procedures/00/00-379-008.html) in Section 5, for more information on interpretation of test results.

Note : Dispose of used coolant or antifreeze in accordance with federal, state, and local laws and regulations.

Cummins Inc. recommends the use of Fleetguard® coolants.

Coolant Filters

Coolant filters are optional on all Cummins equipment. Most Cummins® engines built prior to January 1, 2013 were manufactured with a coolant filter head. Use of a coolant filter is recommended, especially in applications where cleanliness may be an issue. If desired, an aftermarket engine coolant filter head from Cummins Filtration™ can be easily installed on most engines.

Several types of coolant filters are available for use on Cummins® engines. The choice of coolant filter type will depend on choice of coolant formulation and choice of service interval. If coolants capable of an extended service interval are used, an extended service interval coolant filter **must** also be used.

Chemical Filters

Some coolant filters contain additives for Supplemental Coolant Additive (SCA) replenishment. These filters, usually referred to as chemical filters, should **only** be used with conventional and hybrid coolants that require SCA replenishment.

Chemical-Free Filters

OAT coolants do **not** require SCA replenishment and **must** use chemical-free coolant filters.